Reflection Journal: Text Representation in NLP

Completing this lab helped me understand how computers convert human language into numerical representations that machine learning models can process. Before this activity, I understood that computers work with numbers, but I did not fully understand the different methods used to represent text or the information that each method preserves and loses. Through this lab, I learned about text preprocessing, Bag of Words, TF-IDF, n-grams, word embeddings, cosine similarity, and sentiment classification.

What I Learned

The first thing I learned was the importance of text preprocessing. Raw text can contain uppercase letters, punctuation, stop words, and different forms of the same word. These variations can create unnecessary features and make the dataset more difficult to analyze. I practiced converting text to lowercase, removing punctuation, splitting sentences into tokens, removing stop words, and applying stemming. For example, words such as “movies” and “watching” were reduced to simpler forms such as “movi” and “watch.” This creates more consistent input for machine learning models.

I also learned how Bag of Words represents a document by counting how many times each vocabulary word appears. This method is simple and interpretable because each position in the vector corresponds to a specific word. However, it does not preserve word order or context. For example, “the dog ate my homework” and “the homework ate my dog” can receive identical Bag of Words vectors even though they have completely different meanings.

TF-IDF taught me that all words should not be treated as equally important. Common words that appear in many documents receive lower weights, while distinctive words receive higher weights. For example, “movie” may appear in almost every review, but “cinematography” may appear in only one review and provide more useful information about that document. This makes TF-IDF useful for search engines, document ranking, and text classification.

The n-gram section showed me how word sequences can preserve some context. Bigrams and trigrams can represent phrases such as “great movie,” “poor acting,” and “not very good.” These phrases communicate more information than individual words. I also learned about dense word embeddings, which place semantically related words close together in vector space. The most surprising concept was that mathematical operations such as “king − man + woman” can produce a vector close to “queen.” This demonstrated how an embedding space can capture meaningful relationships between words.

Finally, I built sentiment classifiers using Bag of Words, TF-IDF, and n-gram features. All three methods achieved perfect accuracy on the small balanced teaching dataset. This showed me that simple text representations can perform well when the classes contain clear positive and negative

vocabulary. However, I also understood that perfect accuracy on a controlled dataset does not guarantee the same performance on more complicated real-world reviews.

Challenges I Faced and How I Overcame Them

One challenge was correctly implementing the preprocessing steps in the proper order. Tokenization, stop-word removal, and stemming must be applied carefully. Removing an important word such as “not” could change “not good” into “good” and reverse the meaning. I overcame this challenge by testing the preprocessing function with a sample sentence and comparing its output with the expected solution.

Another challenge was manually implementing Bag of Words and TF-IDF. I needed to understand how to build a vocabulary, count each word, calculate term frequency, determine how many documents contained a term, and apply the inverse document frequency formula. I overcame this by dividing each function into smaller steps and checking the results using the solution cells and scikit-learn’s `CountVectorizer` and `TfidfVectorizer`.

The high dimensionality and sparsity of Bag of Words were also challenging concepts. Most values in the document vectors were zero because each review used only a small portion of the full vocabulary. I understood this problem better by examining the heatmap and calculating the sparsity percentage. This helped me see why sparse matrix formats and dimensionality-reduction techniques are important for larger datasets.

I also faced a resource challenge because the NLTK datasets and pretrained GloVe embeddings may require an internet connection. To overcome this, the notebook used resource-safe tokenization and an offline demonstration dataset and embedding model. These alternatives allowed every code cell to run successfully while still demonstrating the main concepts.

Another important issue was dataset balance. The original selection process could have chosen only positive reviews because all positive examples appeared first in the dataset. A model trained on one class would fail to learn meaningful positive-versus-negative differences. I overcame this by selecting equal numbers of positive and negative reviews and shuffling them before splitting the data into training and testing sets.

What I Want to Explore in the Future

In the future, I would like to explore contextual embeddings and transformer models such as BERT. Traditional word embeddings assign one vector to each word, but contextual models can give a word different representations depending on the sentence. For example, the word “bank” has different meanings in “river bank” and “bank account,” and a contextual model can recognize that difference.

I would also like to study subword tokenization, sentence embeddings, semantic search, attention mechanisms, and multilingual NLP. In addition, I want to test these representation methods on larger and more realistic datasets containing sarcasm, mixed opinions, spelling errors, and

informal language. I am also interested in bias detection because embeddings can learn unfair stereotypes from their training data.

Overall, this lab taught me that text representation is not simply about converting words into numbers. Each method captures different information and has different strengths, limitations, and ethical concerns. This experience gave me a stronger foundation for studying modern NLP systems and applying text analysis to real-world problems.